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RESEARCH ARTICLE

Generalized Ansatz for High-Accuracy Quantum Sentiment Classification Using Variational Quantum Algorithm Scheme

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ABSTRACT This paper investigates sentiment classification using QNLP within the Variational Quantum Algorithms (VQA) framework, which offers a practical approach for current hardware limitations by optimizing parameterized quantum circuits through classical computation. While the quantum computer is not ready yet for commercialization, the development and testing of quantum algorithms in classical simulations are useful for finding the quantum advantages. Moreover, designing an effective VQA ansatz is a key challenge because it directly influences the model's expressiveness and trainability. To address this, a new ansatz called the GeneralQC ansatz is proposed by combining the preliminary SimpleSA ansatz with the circuit-centric quantum classifier, aiming to enhance performance by integrating the strengths of both methods. Three instantiations of the GeneralQC ansatz—H-CRx, Rx-CNOT, and Rx-CRz—are explored under the SimpleSA strategy of defining noun word boxes as fixed identity gates and are compared with both the baseline SimpleSA ansatz and versions with noun parameterization. The findings reveal that the Rx-CNOT and Rx-CRz models with non-parameterized nouns significantly outperform the SimpleSA ansatz, achieving an accuracy of 96.92%.

INDEX TERMS Quantum natural language processing, sentiment classification, ansatz.

I. INTRODUCTION

Quantum computing is a branch of computing that utilizes quantum mechanics to perform computation. One paradigm of quantum computing is Universal Quantum Computation, first described by Deutsch as an extension of efforts to develop devices capable of controlling quantum systems [1]. The Universal Quantum Computer, also known as a Quantum Turing Machine (QTM) consists of quantum gates and wires operating on quantum bits or *qubits*, which Deutsch claimed was capable of representing universal states [1], [2]. While it is still an open problem whether a QTM can represent every possible physical state, scientists have begun exploring the uses of possible QTMs [2].

One such use is in quantum machine learning (QML). QML is concerned with the utilization of quantum computers to perform machine learning tasks [3]. The motivation stems from the capability of quantum computers to work with high-dimensional spaces that are computationally expensive for conventional devices. Although the effort to quantify the quantum advantage is still ongoing, the inherent polynomial speedup offered by qubits is a substantial incentive [4]. With the advent of NISQ-era devices, it has become possible to perform QML experiments requiring moderate-scale resources

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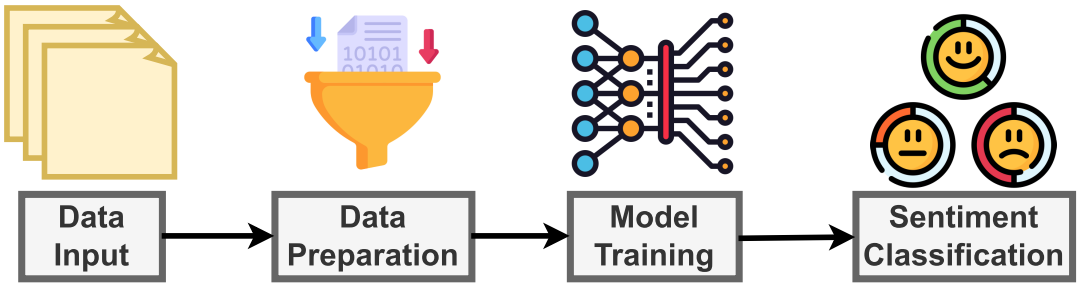


FIGURE 1. Overall illustration of sentiment classification processes.

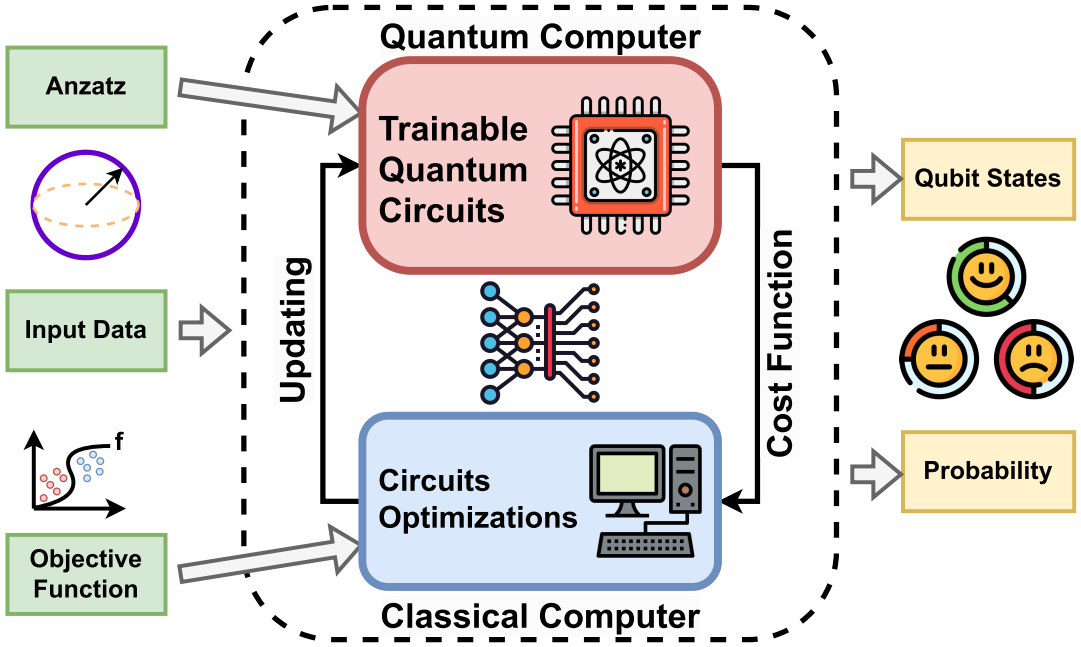


FIGURE 2. Diagram of VQA scheme: An overview of the flow of information in a VQA scheme for sentiment classification.

within the tens of qubits range [5]. One branch of machine learning which has seen significant development over the past few years is natural language processing (NLP), giving rise to the field of quantum natural language processing (QNLP). Fig. 1 shows the overall NLP processes present in a standard NLP task, shown for this case in sentiment classification. As a general machine learning pipeline, the process starts with providing and preparing the data. After that, the data is used for training with specific machine learning algorithm and classification.

The foundations for QNLP were established in [6] and [7]. Their seminal papers introduced the DisCoCat framework, which mapped natural language sentences onto pregroup grammars which were represented by quantum process diagrams. This grammar-preserving approach marked a departure from the bag-of-words models that dominated the NLP scene of the late 2000s and offered a more structured representation of language [8], [9], [10].

The arrival of the NISQ era – the Noisy, Intermediate-Scale Quantum era – would facilitate the limited implementation of QML models in the late-2010s [11]. These quantum devices are characterized by the scale of their systems in the tens-to-several-hundreds of qubits range, termed the "intermediate-scale". Furthermore, these devices are noisy and susceptible to errors during measurement, performing operations, stability, and entanglement. Despite these challenges, the noisy behaviour has been exploited in a certain class of QML models. A hybrid quantum-classical approach known as Variational Quantum Algorithms (VQAs) work around the drawbacks of NISQ devices by delegating the calculation of parameter update to classical computers [12]. An overview of the flow of information in a VQA framework is depicted in Fig. 2. The scheme incorporates both quantum and classical computers to solve a problem *i.e.,* sentiment analysis. First, the input data, objective function, and ansatz are defined in the form of quantum gates. Then, the quantum circuits are trained using the corresponding data, producing qubit states and measurement probabilities as output. Finally, the circuit is optimized by updating the parameters of the quantum gates.

Over the last four years, significant progress has been made in QNLP due to advancements in quantum computing.

In [13], a QNLP pipeline grounded in VQA principles was introduced. The authors also published the Lambeq Python library, which was intended to facilitate independent QNLP implementation. The framework is based on the DisCoCat model, initially proposed in [6]. This pipeline outlines the process of converting input sentences into predictions using quantum machine learning models, as shown in Fig. 3, which is cited from [13].

An NLP task which has received attention for investigation using QNLP is sentiment analysis, a popular feature of natural language understanding. Recent studies [14], [15], [16], [17] have demonstrated the effectiveness of QNLP methodologies in binary sentiment classification tasks, integrating the DisCoCat model with parameterized quantum circuits (PQCs) instantiated via the Instantaneous Quantum Polynomial (IQP) ansatz. These studies draw from [13], which establishes a pipeline for sentence-level classification using QNLP. Notably, [14] introduces a hybrid quantum-classical model for sentiment classification along with basic representation modeling. The hybrid approach is further investigated in [15], which utilizes the quantum-enhanced support vector machine originally proposed in [18]. An example application of the QNLP technique for sentiment classification is performed in [16], which achieves 100% accuracy on an experimental dataset of 130 sentences. An overarching study of the VQA framework QNLP is based on is explored in [17], which proposes a new ansatz tailored for the sentiment classification task and achieves 85.00% accuracy on a more complex dataset of 280 sentences. These results highlight the potential of quantum-inspired frameworks to enhance the accuracy and efficiency of sentiment analysis. Despite promising advancements, QNLP encounters significant optimization challenges, particularly in relation to computational efficiency and model performance. Current methodologies, including the use of Simultaneous Perturbation Stochastic Approximation (SPSA) in classical optimization [13], [14], [17], underscore the necessity for further enhancements. Addressing these challenges is crucial to unlocking the full potential of quantum computing in NLP, motivating ongoing research into novel hybrid quantum-classical strategies.

A critical factor in the performance of VQAs is the choice of ansatz [19]. In quantum computing, an ansatz functions as the model or template for the architecture of a quantum circuit. In the context of QNLP, an ansatz functions as the bridge between the categorial grammar representation and the quantum circuit representation. As categorial grammars can be represented using string diagrams, the ansatz operates by dictating how the components of grammatical string diagrams (strings and word boxes) are transcribed to quantum circuit gate arrangements. In [19], ansätze are broadly classified into two groups: hardware ansätze and problem ansätze. Hardware ansätze are generic and can be employed in a wide range of problems, and they are only optimized by implementation over the set of gates natively available on a quantum hardware [19]. Problem ansätze meanwhile, are not task-agnostic and designed specifically to handle the properties of specific problems.

An effort to design a problem ansatz for sentiment classification has been made in [17]. The proposed SimpleSA ansatz draws its architecture from the hardware ansatz in [18] while employing a parts-of-speech-based masking strategy from [20]. Where in previous ansätze such as IQP ansatz the word boxes representing nouns are instantiated as parameterized SU(2) single qubit gates, the proposed SimpleSA ansatz foregoes this parameterization and implements nouns as fixed identity gates. In other words, the ansatz ignores noun parameterization and instead only instantiates word boxes for compound types such as verbs and adjectives. The modified ansatz achieves an accuracy of 85.00% on the 2022 dataset, compared to the 72.55% of the baseline IQP ansatz. This paper builds upon the noun parameterization strategy and investigates what alternative architectures can be used for quantum sentiment classification.

A framework that highly suits sentiment classification is the Circuit-Centric Quantum Classifier offered in [21]. Circuit-Centric Quantum Classifiers (CCQCs) are notably designed to be low-depth while maintaining a poly-logarithmic number of trainable parameters with respect to the input dimension. This allows circuits to have sufficient expressibility over a limited amount of parameters, and is therefore suitable for binary classification tasks on NISQ devices.

Some recent work on sentiment analysis using quantum circuits is presented in [22], [23], [24], and [25]. In [22], a quantum multimodal neural network is proposed to perform sentiment analysis tasks. It consists of multimodal data preprocessing, unimodal feature extraction, multimodal feature fusion, and optimization. In [23], the multimodal approach is also used to detect sarcasm, sentiment, and emotion in images and sentences. Furthermore, in [24], sarcasm, sentiment, and analysis are also analyzed using a hybrid quantum-classical neural network or using a variational quantum circuit (VQC). Finally, in [25], a pretrained approach is used to improve the performance of the NLP tasks. However, these works mostly did not focus on the selection of an ansatz and its generalization to enhance the performance of the analysis.

The main goal of this research is to explore alternative hybrid quantum-classical machine learning methodologies for sentiment analysis in order to surpass the current benchmarks established in [17]. In this paper, the authors develop and apply novel ansatz algorithms inspired by the CCQC framework on quantum sentiment classification. **Our main contributions** are as follows:

1) We introduce the GeneralQC ansatz, which draws from both the SimpleSA ansatz and the CCQC framework;
2) We expand the QNLP sentiment classification dataset from 280 sentences to 886 sentences;
3) We analyze the performance of several constructions of the GeneralQC ansatz; and
4) We analyze the suitability of the ansatz using coherence, a measure of non-classicality.

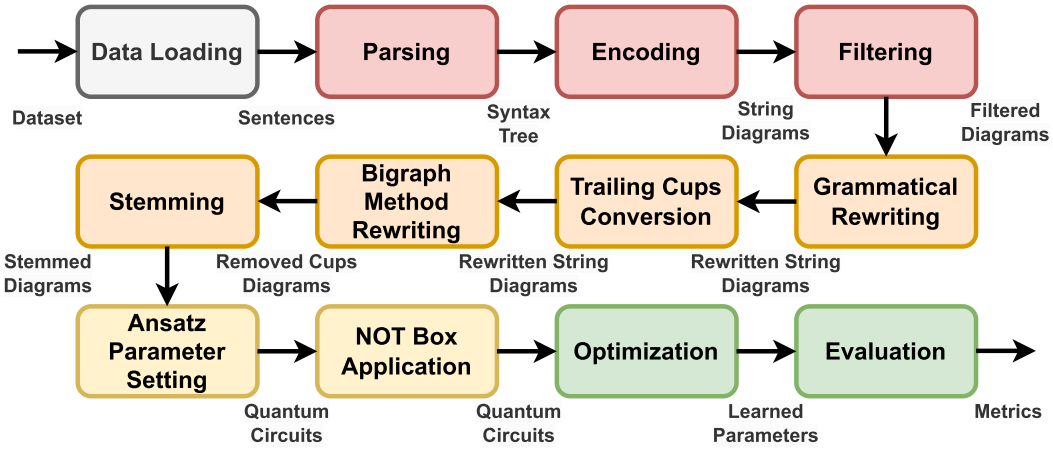


FIGURE 3. QNLP pipeline diagram. The processes are grouped by color based on the corresponding section of the canonical pipeline in [13].

By enhancing the interpretative power and computational efficiency of QNLP models, this research aims to improve the performance of QNLP-based sentiment classification and expand the library of quantum-based machine learning applications in preparation for the post-NISQ era.

II. RELATED WORKS

A. SENTIMENT ANALYSIS

Sentiment analysis (SA) is the subfield of NLP concerned with machine learning techniques to determine the sentiment or approval of an idea conveyed through natural language [8]. In practical implementations, the sentiment of a text is either classified as positive, neutral, negative, or a scaled value denoting the strength of positive or neutral sentiment held by the author of the text. Over the past decade, sentiment analysis has become ubiquitous in business intelligence, where large amounts of text are processed automatically to determine the customers' or users' opinion towards a product, service, or public event [9]. Examples of the source of text are customer reviews, social media posts, or published news articles.

Sentiment analysis tasks are distinguished based on the scope of analysis. The scope may be determined by the length of text, *i.e.,* document-level or sentence-level sentiment analysis which conclude the general positive or negative sentiment held over the body of text, or on an aspect level, which targets certain reviewed features. There are a variety of machine learning methods for performing sentiment analysis. A rudimentary model uses the Naïve Bayes algorithm. Naïve Bayes is capable of accomplishing sentence-level classification tasks that depend on the vocabulary of a text. It has found popular use in spam detection and political bias classification. The algorithm works by examining a sentence and calculating the probability that the words composing the sentence would appear in a positive-sentiment text or a negative-sentiment text [10]. Other than Naïve Bayes, artificial neural networks (ANNs) can also be used for sentiment analysis. Traditionally, the corpus would be pre-processed to determine the numerical embedding of every word, as in a distributional model of semantics. This numerical encoding is then used as the inputs of the neural network. Most applications use a Recursive Neural Network (RNN), which has been found to be most suitable for sentiment classification due to its mechanism that processes information sequentially.

More recently, models equipped with attention mechanisms known as transformers have been employed widely for natural language tasks. The transformer model is composed of an encoder and a decoder, which sequentially receive embedded word tokens as input. The decoder is fed the output embedding of the encoder to yield the probabilities of output tokens [26]. Thus, the transformer model can be adapted for a variety of natural language tasks, including sentiment classification [27]. Among the state-of-the-art transformer architectures are BERT, XLNet, ALBERT, GPT2, and the text-to-text transfer transformer (T5). A comprehensive review [27] found that transformer models are more capable of recognizing sentiments that involve informal language (*e.g.,* sarcasm), as well as scoring higher on datasets with large vocabularies. Among notable transformer architectures, the T5 model is shown to outperform the rest on multiple sentiment analysis datasets [27].

Nevertheless, neither method considers the syntactic structure of a sentence. That is, Naïve Bayes views sentences as a bag of words, RNNs' capability to process sequential inputs are limited to windows of set lengths, and transformer models only rely on positional embedding. In reality, fine-grained sentiment analysis requires the consideration of grammatical structure [28]. Furthermore, different parts-of-speech influence the sentiment of a sentence at different levels [20]. The paper investigates the performance of SA models given different rankings of each parts-of-speech, obtaining the best results when nouns are ranked least important and verbs and adjectives are ranked most important. This highlights a gap in previous methods that overlooked the explicit role of parts-of-speech or lexical categories in

shaping sentiment. This drawback of probabilistic models explains how the compositional structure of DisCoCat QNLP allows the resulting models to be interpretable.

B. QUANTUM NATURAL LANGUAGE PROCESSING

The framework used, known as DisCoCat, originates from [6]. The name is an abbreviation of ''distributional'', ''compositional'', and ''categorial'', indicating the features of the framework. It uses categorial grammar to supply distributional semantics with compositional properties. That is, it employs Lambek grammar to represent sentences, then defines a mapping from pregroup types to a vector space. The meaning space follows under Lambek calculus, allowing sentences to be manipulated via composition to construct complex sentences, evaluate semantics, or deconstruct sentences into phrases. Two commonly defined grammatical types are n and s, representing the noun type and sentence type respectively.

Sentences represented using this grammar may be depicted using string diagrams, which consist of boxes and strings [29]. Boxes depict words, whereas strings depict relations between grammatical types. Formally, strings depict the state of objects, and boxes depict morphisms that transform objects [6]. The string diagrams may be manipulated using a set of graphical calculus rules that conserve the semantics of the sentence being represented. The diagrams are interpreted by evaluating the propagation of information from word boxes to other word boxes through the strings that connect them. The meaning value of a sentence is obtained at the singular open s string [6].

C. HYBRID QUANTUM-CLASSICAL MACHINE LEARNING

Variational Quantum Algorithms (VQAs) are a framework of hybrid quantum-classical machine learning which define machine learning models as trainable circuits. The training process starts with an approximation (known as an ansatz) and is optimized gradually to fit the target states, akin to the variational method [12]. The variational circuits are evaluated on quantum computers, and the outputs are passed on to classical computers. The optimization process is delegated to the classical computer using conventional optimization algorithms, and the updated parameters are passed back to the quantum computer. Formally, the VQA optimization task can be defined as iterating optimization to find the values of θ that minimize a cost function $C(\theta)$. The solution is commonly understood as a point where $C(\theta)$ is a minima, that is, $\frac{\partial C(\theta)}{\partial \theta} = 0$. Explicitly, the objective function is described as $C(\theta) = f(\rho, O, U(\theta))$ where ρ is the initial input state, O is a set of observables (usually the Pauli-Z basis), $U(\theta)$ is the hybrid quantum-classical machine learning (HQCML) model described by a circuit unitary operator, and Tr is the trace function. It is calculated as the sum of objective scores at each observation point k as in (1) [12].

$$C(\theta) = \sum_k f_k(Tr[O_k U(\theta)\rho_k U^\dagger(\theta)]) \tag{1}$$

An important component in VQA is the ansatz used [19]. The ansatz dictates how the problem is converted from a descriptive form into its quantum representation. Ansätze can be understood as the generic architecture that is used to build quantum machine learning models, such that the circuits can be trained to obtain an accurate model [30]. Generally, VQA ansätze have the form in Eq. (2) [12].

$$U(\theta) = U_L(\theta_L)\ldots U_2(\theta_2)U_1(\theta_1) \tag{2}$$

$$U_l(\theta_l) = \prod_m e^{i\theta_m H_m} W_m \tag{3}$$

where $U_l(\theta_l)$ in Eq. (3) is the unitary dependent on θ_L, the l-th element of θ, H_m is some Hermitian operator acting on the system, and W_m is some fixed (independent of any parameters) gate forming the non-trainable section of the ansatz.

D. IQP ANSATZ

An ansatz commonly used in QNLP is the Instantaneous Quantum Polytime (IQP) ansatz. It is used in [13] in one of the first practical demonstrations of QNLP. While IQP itself is a broad class of quantum circuits, the ansatz presented in [13] is implemented as an architecture consisting of layers. Each layer or block has a row of Hadamard gates, followed by an entangling layer made of controlled unitary rotation gates. The entangling layer may be linear, that is, it entangles every pair $i, i+1$ in the register, or fully entangled between every pair i, j.

Following the example in [13], the IQP ansatz is employed in [14]. In [14], the QNLP pipeline from [13] is expanded and further investigated by exploring suitable representations for sentiment analysis. Notably, it models the word ''not'' as the Pauli-X fixed quantum gate placed at the output s string.

E. SIMPLE SENTIMENT ANALYSIS ANSATZ

The Simple Sentiment Analysis (SimpleSA) ansatz is an ansatz designed for quantum sentiment classification that was proposed in [17]. The ansatz employs a part-of-speech based strategy by treating noun word boxes as identity transformations. In other words, the ansatz does not instantiate nouns as a single qubit unitary operator, but rather as an identity operator. This results in a quantum circuit whose readout is not affected by noun parameters. The entangling layer is likewise adjusted to maintain expressibility of the resulting circuits by utilizing a CNOT-Rz-H construction. The SimpleSA template for a compound type construction is depicted in Fig. 4.

The paper reports a significant increase in performance. Where the IQP ansatz results in 72.55% accuracy, the SimpleSA ansatz achieves 85.00% accuracy [17].

F. CIRCUIT-CENTRIC QUANTUM CLASSIFIERS

Circuit-centric quantum classifiers are a class of HQCML classifiers proposed in [21]. The framework consists of a set of quantum circuits known as the model circuits, that

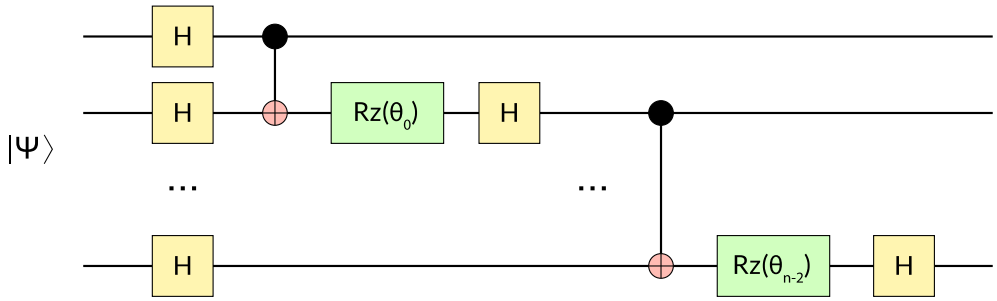


FIGURE 4. SimpleSA Ansatz construction. The circuit consists of an H-gate layer followed by cascading CNOT-Rz-H blocks.

is, a set of circuits which are used as part of the inference process within a machine learning model. The complete inference process is divided into four main steps. First, a classical dataset $x = \{x_0, \ldots, x_{n-1}\}$ is encoded onto quantum circuits using some non-linear map $\varphi(x)$. The prepared states are then passed through the model circuit composed of a parameterized unitary U, obtaining a final state $|\varphi'\rangle$. Measurement is then performed to obtain the probability distribution of the output qubit $|\langle i|\varphi'\rangle|$. Lastly, values are post-processed by the introduction of a trainable bias value and thresholding. The end-to-end process yields the model inference as Eq. (4).

$$f(x;\theta,b) = \begin{cases} 1, & \text{if } \Sigma_i^n |\langle 1\rangle|(U_\theta\varphi(x))_i| + b \geq 0.5 \\ 0, & \text{otherwise} \end{cases} \quad (4)$$

The general gates G may be instantiated with groups of common universal quantum gates such as parametric rotation gates or Hadamard gates. In this paper, ansätze with structures based on circuit-centric quantum classifiers are referred to as GeneralQC ansatz.

The Sim15 ansatz is a specific instantiation of the GeneralQC ansatz with tensored *Ry* layer and entangling *CNOT* layer [31].

G. COHERENCE

The coherence of a quantum system is the degree of superposition that the system is in, with respect to some set of basis states. A coherent system is described in contrast to an incoherent system, that is, a system composed only of basis states. An incoherent system is also known as classical, since it has no mixing properties. Hence, the term non-classicality can be used interchangeably with coherence. Formally, a state $|\Phi\rangle$ is incoherent if its density matrix can be written in the form of Eq. (5) for a set of orthonormal bases $\{|a\rangle\}$. As conventionally the computational basis is used, therefore an informal interpretation of the term coherence is the degree of superposition between computational bases. However, a state can be coherent in one basis and incoherent in another. Therefore, the term maximum coherence refers to the measure of coherence over all possible basis states [32].

$$|\Phi\rangle\langle\Phi| = \Sigma_a p_a |a\rangle\langle a| \quad (5)$$

There have been many definitions of coherence measurement to suit varying tasks and contexts [33]. In [32], coherence is measured through non-classicality, where non-classicality *NCl* can be calculated in Eq. (6) using the intermediate measure $P(a,b|\rho)$. The intermediate measure is defined in Eq. (7), where a, b are sets of orthonormal bases $\{|a\rangle\}$, $\{|b\rangle\}$, ρ is the quantum state of interest, Π is the density matrix or projection operator, and $Tr()$ is the trace operator.

$$NCl(P(a,b|\rho)) = \Sigma_{a,b}|(a,b|\rho)| - 1 \quad (6)$$

$$P(a,b|\rho) = Tr(\Pi_a\Pi_b\rho) \quad (7)$$

This value is used to calculate coherence, which is defined in Eq. (8).

$$C(\rho;\{\Pi_a\}) = max_{\{|b\rangle\}} NCl(P(a,b|\rho)) \quad (8)$$

H. NOISE AND ERROR MITIGATIONS IN QC

As circuit depth increases, the impact of noise on both convergence and computational results becomes more significant. In this subsection, we briefly describe several quantum error correction (QEC) methods used to mitigate noise in quantum computers.

Quantum error correction (QEC) is a promising strategy for addressing noise-related challenges in quantum computing. One class of QEC codes enhances error detection by using additional flag qubits. These codes employ low-degree graphs to reduce frequency collisions and improve performance in superconducting qubit systems. The approach includes subsystem codes on a "heavy-hexagon" lattice, combining features of surface and Bacon-Shor codes, and modified surface codes on a "heavy-square" lattice.

Another method realizes a hardware-efficient logical qubit memory using concatenated bosonic codes arranged in a linear array. In this architecture, bit-flip errors are naturally suppressed using stabilized cat qubits, while phase-flip errors are corrected with a repetition code, benefiting from noise bias. This design achieves high error thresholds and efficient scaling with the physical qubit count.

Furthermore, researchers have demonstrated the experimental realization of entangled logical qubits (ELQs) and their protection using QEC in a bosonic quantum module. The

logical encoding in a high-dimensional space allows errors to be corrected and entanglement to be preserved despite environmental noise. Purification further enhances fidelity, highlighting bosonic modules as promising building blocks for scalable quantum processors and networks.

III. PROPOSED ANSATZ: GENERAL QUANTUM CLASSIFIER ANSATZ

This paper builds upon the work in [17], namely the SimpleSA ansatz, and improves the ansatz that is tailored for sentiment classification. Specifically, the linearly entangling CNOT-Rz-H layer is replaced with the model circuit defined in [21]. This aims to maintain sufficient expressibility while limiting circuit depth with respect to register size.

A. DATA DESCRIPTION

The dataset used in this research expands the dataset of sentiment sentences used in [14] and [17]. The full data can be accessed at https://github.com/abiwardani/qnlp-sa-fp. This dataset consists of 886 English sentences, with a distribution of 532 training sentences, 160 validation sentences, and 194 testing sentences. The vocabulary is comprised of 42 words: 12 nouns, 8 verbs, 22 adjectives. The detail of sentence types is presented in Table 1. This is a notable increase to the previous experimental dataset of 280 sentences and the original demonstration dataset of 130 sentences.

In the dataset, sentences with negative sentiment are labeled 0, while sentences with positive sentiment are labeled 1. In order to conform the data to the single-qubit outputs of a quantum circuit, the labels are mapped onto the Pauli-Z (computational) bases, *i.e.,* $|0\rangle$ and $|1\rangle$. The chosen convention is to map positive (1) sentiments to $|0\rangle$ or $\left[\begin{smallmatrix}1\\0\end{smallmatrix}\right]$ and to map negative (0) sentiments to $|1\rangle$ or $\left[\begin{smallmatrix}0\\1\end{smallmatrix}\right]$.

There are 92, 28, and 28 negation sentences in the training, development, and testing dataset respectively. Negation sentences are sentences containing the word ''not''. In this dataset, they appear in the form of SX[*not*]A sentences. In other words, the word ''not'' is always attached to an adjective (*e.g.,* ''the staff was not kind''). Furthermore, the shortest sentence length in the dataset are in the form of SVO and SXA sentences at 4 words, and the longest sentence length in the dataset are in the form of SVAO and SX[*not*]A sentences at 5 words.

B. QNLP PIPELINE

The method for converting sentences into trainable quantum circuits is adapted from the pipelines established in [13], [14] and [17]. The QNLP pipeline is depicted in Fig. 3. First, the dataset files are imported into Python. The dataset is previously divided into three: the training set, development, and testing set. Learning is performed on the training set where the development set is used to monitor fitting on unseen data. Final metric evaluation is done on the testing set. The dataset are pre-distributed to ensure that sufficient vocabulary is seen in the training set, thereby eliminating the possibility that the development and testing sets contain entirely new words, as this is outside the capability of the DisCoCat framework. This methodology utilizes logical NOT-box modeling from [14], which instantiates word boxes for the word ''not'' as NOT-gates. To this end, the word ''not'' is temporarily deleted from the sentences. This is to avoid the instantiation of the word box for ''not'' in the parsing process, which will simplify the process of appending the NOT-box to the resulting circuits. Then, the sentences are parsed using the Bobcat Parser from the Lambeq library, resulting in raw string diagrams consisting of n and s grammar types [7]. A simple check is performed to ensure that the resulting string diagrams are valid, *i.e.,* they are reducible to the s (sentence) type. Diagrams that fail the test are removed from the dataset.

After filtering, the valid diagrams are rewritten using the grammatical rewrite rules provided by Lambeq. Rewrite rules perform find-and-replace operations on a defined template. For this pipeline, the auxiliary, connector, determiner, postadverb, preadverb, and prepositional phrase rules are applied. Details regarding the default rewrite rules are available in the Lambeq 0.4.1 documentation. In short, the grammatical rewrite rules simplify diagrams based on the expected operational value of a certain grammatical class. For example, determiners such as the words ''the'' or ''a'' are not expected to alter the semantic value of a sentence, so the rewrite rule models them as the identity operator. This results in more concise string diagrams. The rewritten diagrams are then transformed into normal form via the trailing cups conversion algorithm defined in [14]. In this form, the bigraph method can be applied to the string diagrams to effectively remove cups and caps according to the yanking equations [6]. This process approximately halves the amount of qubit wires present in the diagram. Then, stemming is performed on the word boxes to obtain the stemmed forms, mapping words with different conjugation, modification, or pluralized forms onto the base stems.

After obtaining the stemmed diagrams, the string diagrams are converted into parameterized quantum circuits (PQCs). Parameterization is performed using an ansatz. Unlike previous works, this paper uses a CCQC-based ansatz, coined the GeneralQC (General Quantum Classifier) ansatz. The ansatz used is explained in greater detail in Section III-D. Upon instantiating the PQCs using the GeneralQC ansatz, the NOT-box is applied to sentences which originally contained the word ''not''. The Pauli-X gate is appended to the open qubit wire (the output string), as demonstrated in [14].

The final quantum circuits are then compiled into a set of training functions. That is, each set of training, development, and testing circuits are compiled into their respective sets of prediction functions and cost functions. The prediction function takes a set of parameters as its input and outputs an array of predictions, that is, a list of α^2, β^2 for each measured average circuit output $\alpha\,|0\rangle + \beta\,|1\rangle$. This denotes the probabilities of measuring $|0\rangle$ and $|1\rangle$ for each sentence in the respective dataset. The cost functions meanwhile calculate the binary cross-entropy loss over all predictions in the

TABLE 1. Distribution of types of sentences within the dataset.

Type	Component	Train data	Dev data	Test data	Example
SVO	Subject + Verb + Object	14	4	6	I hated the cafe.
SVAO	Subject + Verb + Adjective + Object	338	98	130	I loved the delicious meals.
SXA	Subject + "to be" + Adjective	180	58	58	The foodservice was fast.

associated dataset. These training functions are utilized by an SPSA optimizer to learn the circuit parameters. An inference batch over the test set is used to measure the metrics for evaluating the model.

C. EXPERIMENTAL ENVIRONMENT

The final training and evaluation pipeline is implemented in Python 3.11.6 using Lambeq 0.4.1 and structured as a Jupyter notebook file. Quantum circuits are programmed using Qiskit 0.42.1 and Pytket 1.15.0. The quantum circuits are run locally using Qiskit's Aer Simulator on an i9 Intel processor PC with a 64GB DDR4 Ubuntu machine. All associated code can be found at https://github.com/abiwardani/qnlp-sa-fp.

D. GeneralQC ANSATZ

As stated in Section III-B, the GeneralQC ansatz combines the noun non-parameterization strategy of the SimpleSA ansatz with the circuit architecture of CCQCs. Drawing from [21], the structure consists of two code blocks. Each code block consists of a layer of single qubit gates and a layer of linearly entangling 2-qubit gates. The structure of the entangling layer is dictated by a control range r, which defines that for every control qubit i in an n-qubit register, it is entangled with the target qubit $i - r$ (modulo n). A template architecture is depicted in Fig. 5 where the horizontal wires represent the qubits where operations are applied from left to right. U_1 and U_2 may be fixed gates or single-parameter gates applied on the qubit. The U_2 gates are controlled gates, where the control qubit is represented by a filled black circle connected to the target qubit by a vertical line.

The generic gates $G = G(\alpha, \beta, \gamma)$ in [21] have been reduced to single-parameter gates to convey the idea that in practice, implementation of the ansatz will involve design choice in selecting what single-parameter gates are used in place of U_1 and U_2. For experimentation in this paper, three constructions are investigated: i) H-CRx, ii) Rx-CNOT, and iii) Rx-CRz. The constructions are written as a single-qubit gate hyphenated with the controlled single-qubit gate. The H-CRx construction is chosen as the CCQC implementation of the IQP Ansatz, whereas the Rx-CNOT and Rx-CRz constructions are inspired by the classifier circuits in [31]. The selection of gates is explicitly limited to Z and X rotation gates because the scope of sentiment analysis investigated in this work is restricted to binary classification.

The template architecture for the GeneralQC Ansatz with H-CRx blocks is depicted in Fig. 6. Each block consists of a layer of n Hadamard gates and a layer of n *CRx* gates. As only the entangling layer is parameterized, this ansatz uses a total of n parameters.

Likewise, the template architecture for the Rx-CNOT construction is depicted in Fig. 7. As with the H-CRx construction, the Rx-CNOT ansatz has two blocks, each consisting of a single-qubit gate layer and a controlled gate layer. Unlike the H-CRx construction however, this ansatz uses parameterized gates (the *Rx* gate) in the single-qubit layer, and uses fixed gates (the *CNOT* gate) in the entangling layer. As only the single-qubit gate layer is parameterized, this ansatz uses a total of n parameters.

Lastly, the template architecture for the Rx-CRz construction is depicted in Fig. 8. This ansatz implementation has tensored *Rx* gates and a linearly entangling layer of *Rz* gates. In this way, both layers in the Rx-CRz construction are parameterized. As such, this ansatz has twice the number of parameters of the H-CRx and Rx-CNOT ansatz for a total of $2n$ parameters.

E. EVALUATION METRICS

In total, there are seven metrics recorded in this experiment. Two are recorded during the training phase, and five are recorded during the evaluation phase. The metrics recorded over the training phase are training accuracy and fit time. The training accuracy is measured as the number of correct predictions in the training set over the total number of training data. The fit time is the total measured time required to train the model over 800 iterations.

Meanwhile, the metrics recorded over the evaluation phase are testing accuracy, precision, recall, F1 score, and coherence. Testing accuracy is measured as the number of correct predictions in the testing set over the total number of testing data. Precision is measured as the number of correct positive sentiment predictions in the testing set divided by the total number of positive sentiment predictions in the testing set. Recall is measured as the number of correct positive sentiment predictions in the testing set over the total number of positive sentiment sentences in the testing data. The F1 score is calculated according to Eq. (9).

$$F1 = \frac{2 \times Precision \times Recall}{Precision + Recall} \tag{9}$$

Lastly, coherence is calculated over the testing set using the measure defined in [32]. Coherence is the metric chosen to quantify the non-classicality of the resulting circuits. In this experiment, coherence is used as an indicator of how well a state is represented as a quantum state, as opposed to a classical state. Therefore, this metric denotes how well the results could be replicated using classical machine learning methods and reflects the necessity of quantum states. The method of calculation is explained further in Section III-F.

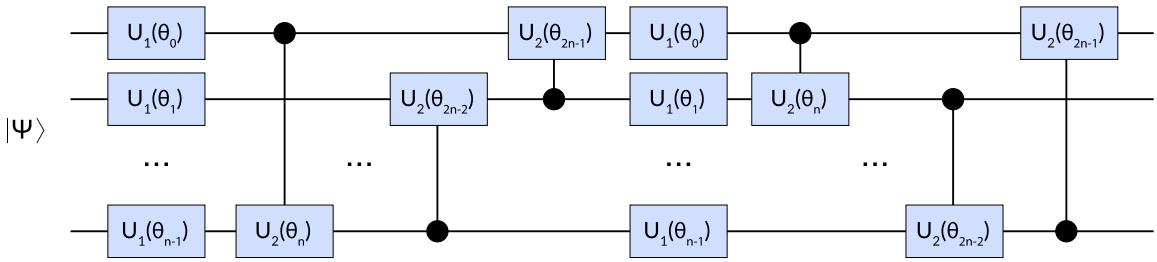


FIGURE 5. GeneralQC Ansatz construction. The circuit consists of two blocks, each block consisting of a single-qubit gate layer and an entangling layer.

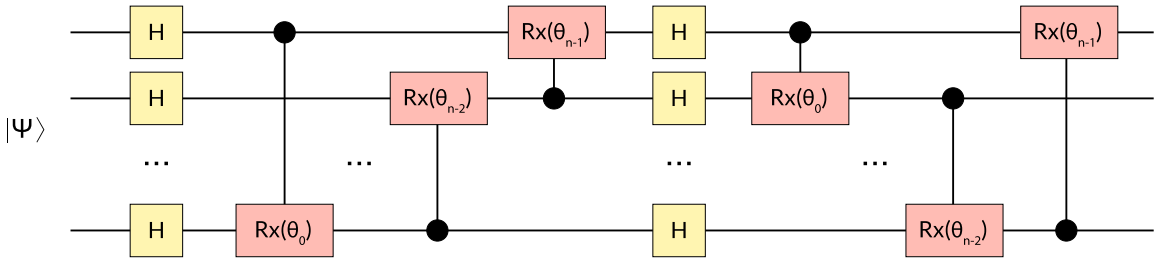


FIGURE 6. GeneralQC Ansatz H-CRx construction. The single-qubit layer uses an array of H gates while the entangling layer uses controlled-Rx gates.

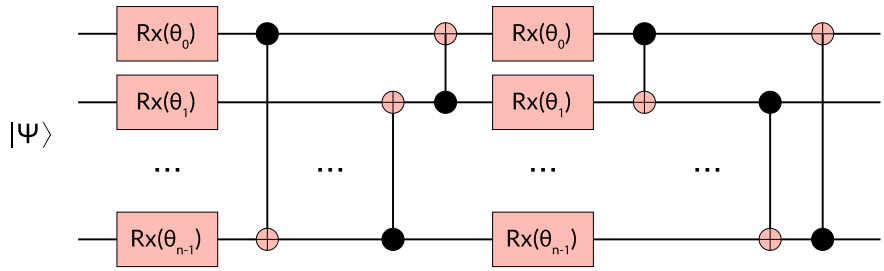


FIGURE 7. GeneralQC Ansatz Rx-CNOT construction. The single-qubit layer uses an array of Rx gates while the entangling layer uses controlled-NOT gates.

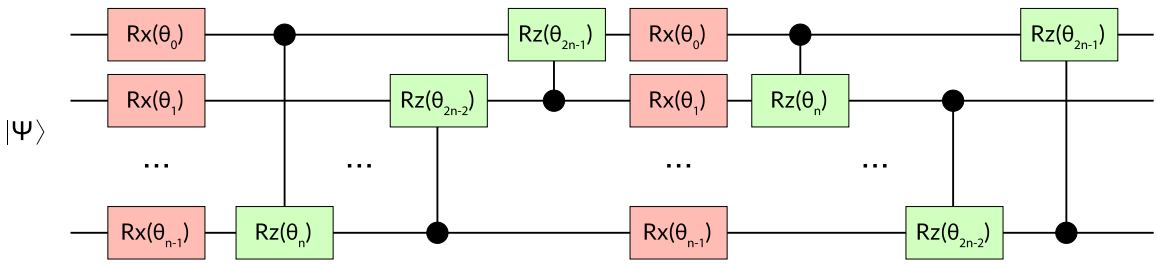


FIGURE 8. GeneralQC Ansatz Rx-CRz construction. The single-qubit layer uses an array of Rx gates while the entangling layer uses controlled-Rz gates.

F. COHERENCE MEASUREMENT

The following describes in detail how the average coherence of a model is measured. Coherence calculation takes place after optimization when the circuit gate parameters have been learned. The process takes the quantum circuits of each test sentence and removes post-selection as well as Z-basis measurement from the circuit, resulting in wires with open ends. An example of such a circuit can be seen in Fig. 9.

This allows quantum simulators to measure the state of the circuit after information has propagated through all gates, also known as the statevectors. The statevector is the vector form of the tensor product of each consecutive qubit state. As convention, this study writes the tensors of qubit states using Big Endian. This is in contrast with the convention used in Qiskit. This is intended to promote readability and understanding within this report. As an example, the statevector of a quantum circuit with four qubits of states $|A\rangle, |B\rangle, |C\rangle, |D\rangle$ is written as $|\Psi\rangle = |A\rangle \otimes |B\rangle \otimes |C\rangle \otimes |D\rangle$. This aligns with the qubit ordering used in circuit diagrams throughout this report.

Upon obtaining the statevector of a test state $|\Psi\rangle$, the value of P associated with Ψ is calculated using Eq. (7). As the formula uses the density matrix, the density matrix of $|\Psi\rangle$ or ρ_Ψ must first be calculated. The density matrix can be determined using the density matrix operator. This yields $\rho_\Psi = |\Psi\rangle\langle\Psi|$, where $|\Psi\rangle$ is the previously obtained statevector of the test circuit.

Slightly jumping ahead to view the calculation of NCl coherence, the formula used in Eq. (8) is defined in reference to two sets of orthonormal bases. For simplicity, this report selects one set of bases to be the computational basis. Reference [32] defines the other set to be the parametric $|b_+\rangle$ and $|b_-\rangle$ states with respect to the rotation angle α and the phase angle β. Maximum coherence therefore maximizes the NCl coherence calculation over all possible $|b_+\rangle, |b_-\rangle$ bases, by searching through all α, β values. However, this approach is unsuitable for this report as test circuit sizes reach four qubits in execution. Searching through all α, β values for four qubits at, for example, a granularity of 100 partitions would require 100^4 iterations.

However as circuits are of maximum 4 qubits in size, calculation dependent on a parametric tensored bloch sphere composition iterating over a range of angle values would be too expensive. As a shortcut, this study uses the multiqubit Clifford group as 2-design unitaries to sample the complete basis space. Consequently, this reduces the complexity necessary from $O(nm^{2q})$ to $O(2^q)$. Asides from complexity reduction, this also escapes the granularity issue present in parametric angle search. To achieve this, the program prepares a Haar-random set of unitaries using the Clifford set of an n-qubit circuit, and takes its density matrix for the projector Π_b in Eq. (7). The Haar sampling is used because maximum coherence calculation attempts to search the value of a function over all possible states, and t-designs are known to perfectly capture any transformation over all possible states to degree t. As basis-coherence values form a 2-subset pair, it is sufficient to use 2-design unitaries, and it is known that the set of all possible states obtainable using only the Clifford group (H and S gates) is 2-design. The average coherence is calculated as the average of maximum coherence over all test circuits. Lastly, in order to measure the relationship between coherence and model performance, the Pearson correlation coefficient between average coherence and test accuracies are calculated.

IV. RESULTS

The results of the experimentation are summarized in Table 2. At 800 iterations, all GeneralQC models outperform or perform equal to the IQP baseline. The Rx-CNOT model outperforms SimpleSA with 87.63% accuracy over SimpleSA's 83.51%, whereas both the H-CRx and Rx-CRz models perform slightly worse on the test set, achieving 82.47% and 79.12% accuracy respectively. However, after removing the nouns, there is a significant improvement in the performance of Rx-CNOT and Rx-CRz. That is, the Rx-CNOT and Rx-CRz models with no nouns vastly outperform SimpleSA, with both models achieving 96. 91% accuracy in the expanded test set, displaying a 13-point increase over the benchmark model and a 17-point increase over the IQP baseline model, as well as outperforming previous performance on smaller test sets reported in [14] and [17]. Meanwhile, the H-CRx model without nouns performs slightly worse on the test set and significantly worse on the training set. However, all GeneralQC ansatze (with or without noun parameterization) require larger resources to train compared to the SimpleSA ansatz, as observed by the large increase in fit times. In terms of computation time over the Aer simulator, SimpleSA maintains a 17-hour lead above GeneralQC models. In the worst case, the Rx-CRz construction with noun parameterization took more than 112 hours to train, taking more than 80 hours longer than SimpleSA. This can be explained in terms of the number of trainable parameters. The SimpleSA model is parameterized by 58 parameters, while the smallest GeneralQC ansatz uses 109 parameters (seen in both the H-CRx and Rx-CNOT models). The Rx-CRz models, however, take significantly longer at 112 hours and almost 89 hours for the models with nouns parameterized and non-parameterized respectively. This is in part due to the fact that the Rx-CRz model has two trainable layers, *i.e.,* the *Rx* section and the *CRz* section, resulting in a total of 218 parameters. Additionally, the largest Rx-CRz circuit (observed in SVAO sentences) has 20 rotation gates, which is over 6 times longer than the largest SimpleSA circuit (3 rotation gates) and twice as long as the largest H-CRx circuit (10 rotation gates). These long fit times show that GeneralQC NLP is currently unsuitable for practical machine learning using simulators and will require real quantum devices to make training times feasibly short. The values for accuracy, precision, recall, and F1 are recorded in Table 3.

From Table 3, the Rx-CNOT and Rx-CRz models have the best accuracy and F1 scores. From the sentence type breakdown in Table 4, the Rx-CRz model has better performance on SXA sentences than the Rx-CNOT model, whereas the Rx-CNOT model has better performance on SVAO sentences. In fact, the Rx-CRz model has 100% accuracy on SXA sentences in training, development, and testing data sets. On SVAO sentences, the Rx-CRz model has lower accuracy compared to Rx-CNOT on all three datasets, but maintains a 100% recall rate. This indicates that it makes no false negatives, and implies that the Rx-CRz model has

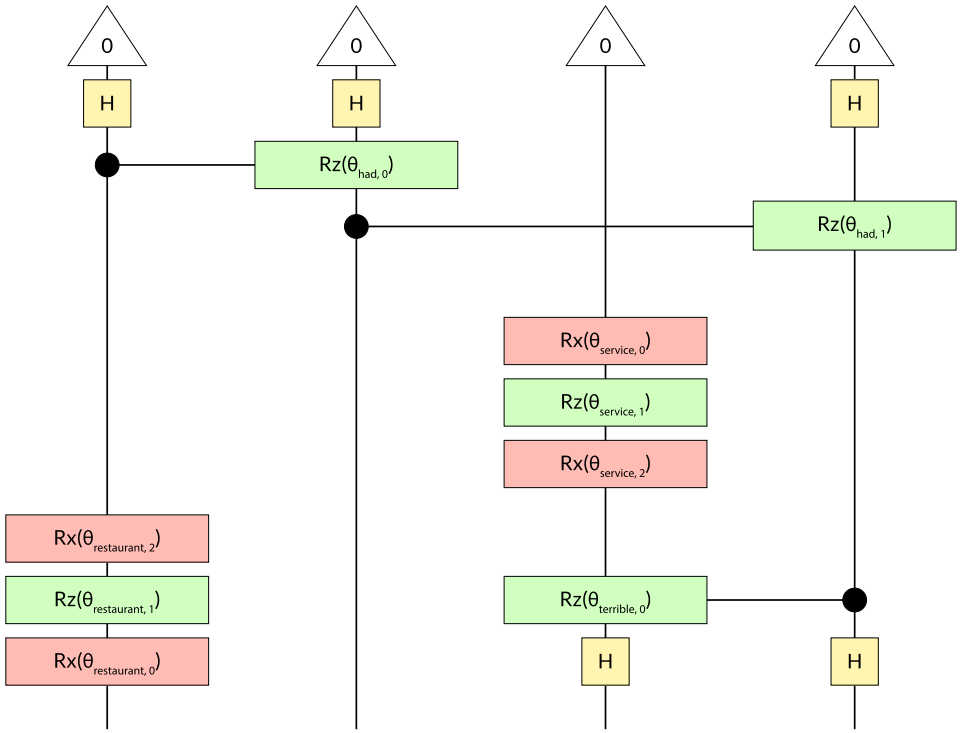


FIGURE 9. Circuit with open wires. Coherence calculation uses the resulting statevector of the system.

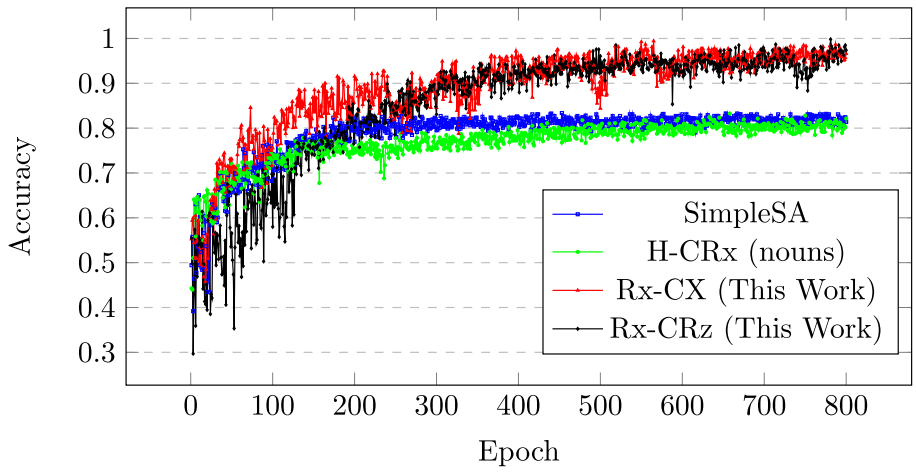


FIGURE 10. Comparison of training accuracies for SimpleSA, H-CRx with noun parameterization, Rx-CX, and RX-CRz.

a bias to predicting positives (as a reminder, in this work assigned to $|0\rangle$). The Rx-CNOT model meanwhile has a higher accuracy on SVAO sentences.

With regards to coherence, it is observed that the Rx-CRz models, both with and without noun parameterization, have the lowest average coherence values, indicating that the Rx-CRz circuits are more incoherent (that is, closer to a classical mixture of orthonormal bases) compared to the other models. However overall, there is a moderate positive correlation between coherence and test accuracy ($r = 0.31$), indicating that models with higher accuracy might require quantum circuits of higher non-classicality.

The graphs of training accuracies and costs of various models are shown in Fig. 10. For the SimpleSA ansatz, the training accuracies are seen to plateau at 80% after about 170 iterations and subsequently oscillate around that value until the 800th iteration. Similarly, the H-CRx (with nouns) model converges around 80% at about 500 iterations, demonstrating ultimately comparable performance while taking longer to train. Meanwhile, the Rx-CNOT (no nouns) and Rx-CRz (no nouns) models train at a much slower rate, reaching around 96% and 98% accuracy respectively at 800 iterations. In fact, the trend for the Rx-CRz (no nouns) model seems to continue upwards, indicating that the model

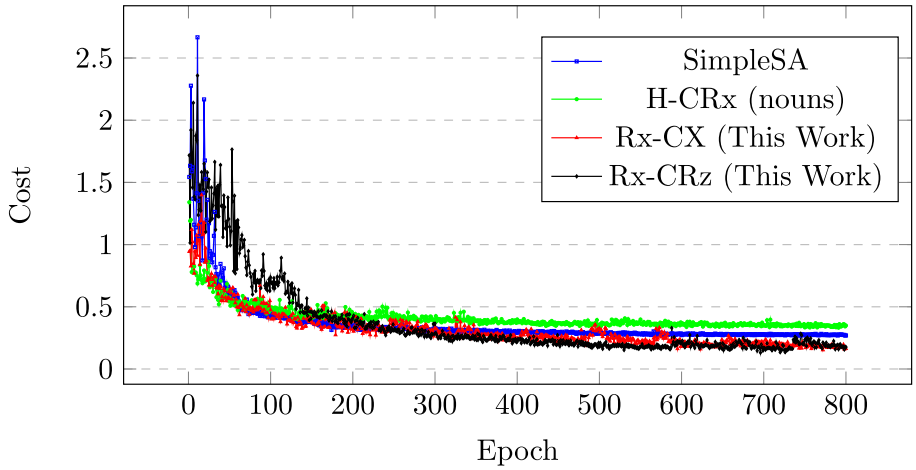


FIGURE 11. Comparison of training costs for SimpleSA, H-CRx with noun parameterization, Rx-CX, and RX-CRz.

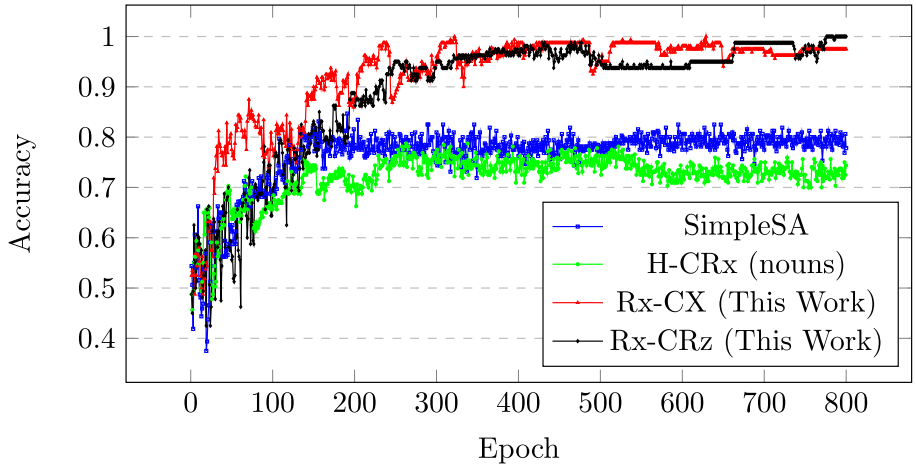


FIGURE 12. Comparison of validation accuracies for SimpleSA, H-CRx with noun parameterization, Rx-CX, and RX-CRz.

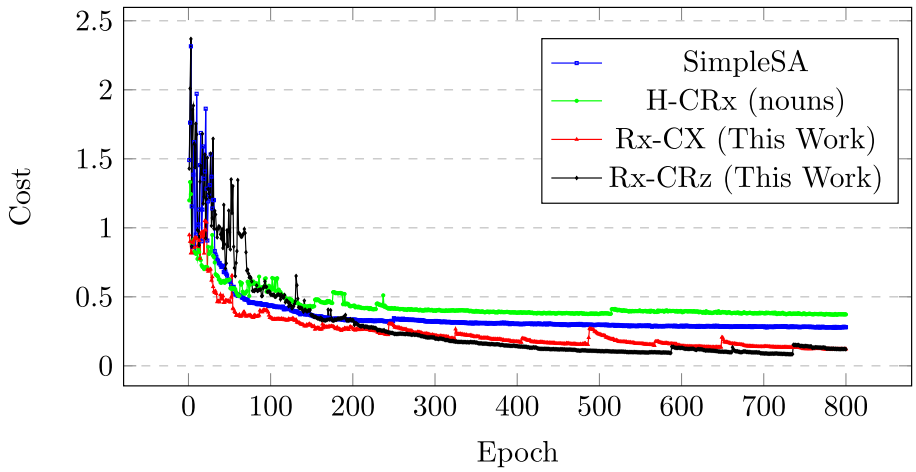


FIGURE 13. Comparsion of validation costs for SimpleSA, H-CRx with noun parameterization, Rx-CX, and RX-CRz.

TABLE 2. Accuracies, fit times, and circuit description of ansatz experimentation. Models are trained on a dataset of 886 sentences with a 532/160/194 training/validation/testing set split. Data preprocessing involves stemming and parts-of-speech annotation for diagram parsing, as explained in Section VI.

Ansatz	Construction	Noun param.	Train acc.	Test acc.	Fit time	No. params	No. rot. gates
IQP [13]	H-CRz	Yes	79.32%	79.38%	73 h 36 min	91	11
SimpleSA [17]	H-CNOT-Rz-H	No	81.39%	83.51%	19 h 59 min	58	6
GeneralQC (This Work)	H-CRx	Yes	82.14%	82.47%	54 h 47 min	142	16
	Rx-CNOT	Yes	90.79%	87.63%	52 h 40 min	142	16
	Rx-CRz	Yes	82.14%	79.38%	112 h 5 min	251	26
	H-CRx	No	78.57%	81.96%	51 h 31 min	109	10
	Rx-CNOT	No	96.62%	**96.91%**	37 h 30 min	109	10
	Rx-CRz	No	97.37%	**96.91%**	88 h 59 min	218	20

TABLE 3. Evaluation metrics of ansatz experimentation.

Ansatz	Construction	Noun param.	Accuracy	Precision	Recall	F1 score	Coherence
IQP [13]	H-CRz	Yes	79.38%	84.15%	71.88%	77.53%	1.399
SimpleSA [17]	H-CNOT-Rz-H	No	83.51%	77.59%	93.75%	84.91%	1.548
GeneralQC (This Work)	H-CRx	Yes	82.47%	79.81%	86.46%	83.00%	1.334
	Rx-CNOT	Yes	87.63%	82.14%	95.83%	88.46%	1.640
	Rx-CRz	Yes	79.38%	76.42%	84.38%	80.20%	1.052
	H-CRx	No	81.96%	77.98%	88.54%	82.93%	1.399
	Rx-CNOT	No	96.91%	95.92%	97.92%	96.91%	1.790
	Rx-CRz	No	96.91%	94.12%	100.00%	96.97%	1.121

TABLE 4. Breakdown of SimpleSA, Rx-CNOT, and Rx-CRz evaluation matrices by sentence type.

Sentence Type	Metric	SimpleSA	Rx-CNOT	Rx-CRz
SVO	Accuracy	100.00%	100.00%	100.00%
	Precision	100.00%	100.00%	100.00%
	Recall	100.00%	100.00%	100.00%
	F1 score	100.00%	100.00%	100.00%
SVAO	Accuracy	70.71%	**96.92%**	95.38%
	Precision	66.96%	**94.29%**	91.67%
	Recall	86.36%	**100.00%**	100.00%
	F1 score	75.34%	**97.06%**	95.65%
SXA	Accuracy	100.00%	96.55%	**100.00%**
	Precision	100.00%	100.00%	**100.00%**
	Recall	100.00%	92.31%	**100.00%**
	F1 score	100.00%	96.00%	**100.00%**

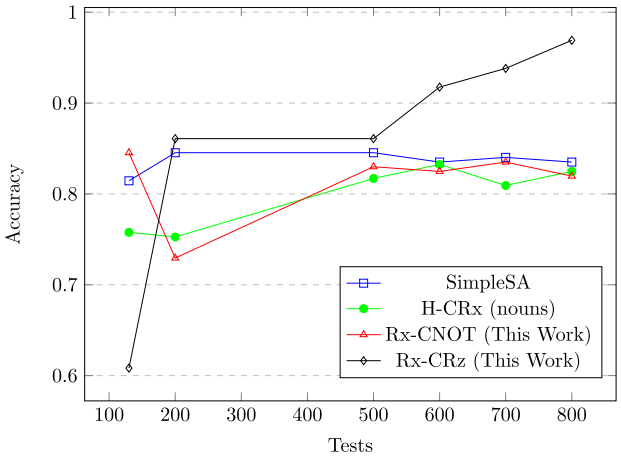


FIGURE 14. Graph of testing accuracies for SimpleSA, H-CRx with nouns parameterization, Rx-CX, and RX-CRz.

still has room for convergence beyond 98% training accuracy after 800 iterations.

V. DISCUSSION

A. SIMPLE SENTIMENT ANALYSIS ANSATZ

The training graph for the SimpleSA ansatz [17] implies that the model is easy to train but has a limited prediction space. That is, it implies that the model has a global minimum at 80% accuracy or that it is difficult for the model to reach the global minimum from the local minima. The simplicity of the model can be demonstrated by deriving the analytical form of the quantum circuit used in the model. Noting that the SimpleSA model has 100.00% accuracy on SVO and SXA sentences and 70.71% accuracy on SVAO sentences, the analysis will be done on a SimpleSA circuit representing a generic SVAO sentence.

To derive the circuit output, the wavefunction must first be obtained. The wavefunction of the circuit is given by Eq. (10).

$$
\begin{aligned}
|\Psi_0\rangle &= |0\rangle^{\otimes 4} = |0000\rangle \\
|\Psi_1\rangle &= H^{\otimes 2} \otimes I \otimes H \, |\Psi_0\rangle \\
|\Psi_2\rangle &= CNOT_{0,1} \otimes I^{\otimes 2} \, |\Psi_1\rangle \\
|\Psi_3\rangle &= I \otimes Rz(\theta_0) \otimes I^{\otimes 2} \, |\Psi_2\rangle \\
|\Psi_4\rangle &= I \otimes H \otimes I^{\otimes 2} \, |\Psi_3\rangle \\
|\Psi_5\rangle &= I \otimes CNOT_{1,3} \, |\Psi_4\rangle \\
|\Psi_6\rangle &= I^{\otimes 3} \otimes Rz(\theta_1) \, |\Psi_5\rangle \\
|\Psi_7\rangle &= I^{\otimes 3} \otimes H \, |\Psi_6\rangle \\
|\Psi_8\rangle &= I^{\otimes 2} \otimes H \otimes I \, |\Psi_7\rangle
\end{aligned}
$$

$$
\begin{aligned}
|\Psi_9\rangle &= I^{\otimes 2} \otimes Rz(\theta_2) \otimes I\,|\Psi_8\rangle \\
|\Psi_{10}\rangle &= I \otimes CNOT_{2,1} \otimes I\,|\Psi_9\rangle \\
|\Psi_{11}\rangle &= I^{\otimes 2} \otimes H^{\otimes 2}\,|\Psi_{10}\rangle
\end{aligned} \tag{10}
$$

Calculating the conditional probabilities uses the probabilities of measuring $|00\rangle$ and $|01\rangle$, which are derived in Eq. (11) and Eq. (12) respectively. The probability of measuring $|0\rangle$ on the second qubit is calculated in Eq. (13). This yields a value of $0.5\cos(\theta_0)+0.5$, indicating that the circuit is indeed simple to train.

$$
\begin{aligned}
P(|0000\rangle) &= |\,\langle 0000|\Psi_{11}\rangle\,| \\
&= |\frac{1}{8}\left(e^{i\theta_0} + e^{i\theta_2} + e^{i(\theta_0+\theta_2)} + 1\right) e^{-\frac{i(\theta_0+\theta_1+\theta_2)}{2}}| \\
&= \frac{1}{64}\,[((\sin(\theta_0) + \sin(\theta_2) + \sin(\theta_0+\theta_2)) \\
&\quad \sin\left(\frac{\theta_0+\theta_1+\theta_2}{2}\right) + (\cos(\theta_0) + \cos(\theta_2) \\
&\quad + \cos(\theta_0+\theta_2) + 1)\cos\left(\frac{\theta_0+\theta_1+\theta_2}{2}\right)\Big)^2 \\
&\quad + ((\sin(\theta_0) + \sin(\theta_2) + \sin(\theta_0+\theta_2)) \\
&\quad \cos\left(\frac{\theta_0+\theta_1+\theta_2}{2}\right) - (\cos(\theta_0) + \cos(\theta_2) \\
&\quad + \cos(\theta_0+\theta_2) + 1)\sin\left(\frac{\theta_0+\theta_1+\theta_2}{2}\right)\Big)^2\Big] \\
&= \frac{1}{16}\,(\cos(\theta_0)\cos(\theta_2) + \cos(\theta_0) + \cos(\theta_2) + 1)
\end{aligned} \tag{11}
$$

$$
\begin{aligned}
P(|0100\rangle) &= |\,\langle 0100|\Psi_{11}\rangle\,| \\
&= |\frac{1}{8}\left(-e^{i\theta_0} + e^{i\theta_2} - e^{i(\theta_0+\theta_2)} + 1\right) e^{-\frac{i(\theta_0+\theta_1+\theta_2)}{2}}| \\
&= \frac{1}{64}\,[((-\sin(\theta_0) + \sin(\theta_2) - \sin(\theta_0+\theta_2)) \\
&\quad \sin\left(\frac{\theta_0+\theta_1+\theta_2}{2}\right) + (-\cos(\theta_0) + \cos(\theta_2) \\
&\quad - \cos(\theta_0+\theta_2) + 1)\cos\left(\frac{\theta_0+\theta_1+\theta_2}{2}\right)\Big)^2 \\
&\quad + ((-\sin(\theta_0) + \sin(\theta_2) - \sin(\theta_0+\theta_2)) \\
&\quad \cos\left(\frac{\theta_0+\theta_1+\theta_2}{2}\right) - (-\cos(\theta_0) + \cos(\theta_2) \\
&\quad - \cos(\theta_0+\theta_2) + 1)\sin\left(\frac{\theta_0+\theta_1+\theta_2}{2}\right)\Big)^2\Big] \\
&= \frac{1}{16}\,(-\cos(\theta_0)\cos(\theta_2) - \cos(\theta_0) + \cos(\theta_2) + 1)
\end{aligned} \tag{12}
$$

$$
\begin{aligned}
P(O=|0\rangle) &= P(|0\rangle_1\,|\,|0\rangle_{0,2,3}) \\
&= \frac{P(|0000\rangle)}{P(|0000\rangle) + P(|0100\rangle)} \\
&= \frac{\frac{1}{16}(\cos(\theta_0)\cos(\theta_2) + \cos(\theta_0) + \cos(\theta_2) + 1)}{\frac{1}{8}(\cos(\theta_2)+1)} \\
&= \frac{1}{2}\frac{(\cos(\theta_0)+1)(\cos(\theta_2)+1)}{\cos(\theta_2)+1} \\
&= 0.5\cos(\theta_0) + 0.5
\end{aligned} \tag{13}
$$

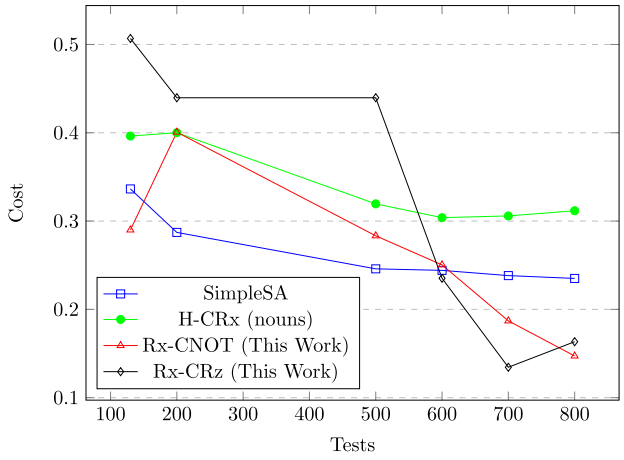


FIGURE 15. Graph of testing costs for SimpleSA, H-CRx with nouns parameterization, Rx-CX, and RX-CRz.

B. H-CRX ANSATZ

As seen in Table 2, the H-CRx model reaches a maximum testing accuracy of 82.47% and 81.96% for the parameterized and non-parameterized noun versions respectively. An inspection of the training graph in Fig. 10 indicates that the model indeed plateaus around the 80% mark. Upon further inspection of the model's inference results, it is found that a majority of the incorrect predictions are cases where the model produces a highly superpositioned state. That is, the model does not strongly predict a $|0\rangle$ or $|1\rangle$, but rather a state composed of nearly $\frac{1}{\sqrt{2}}$ amplitude of each. This can be verified by observing an example of an H-CRx circuit, in this case an SVAO sentence, and substituting the parameters for the verb while allowing the adjective parameters to remain parameterized. This graph is shown in Fig. 16, where the θ_3 and θ_4 parameters representing the adjective are assigned to the x-axis and y-axis respectively. The z-axis meanwhile shows the probability of measuring a $|0\rangle$ for the given parameter values. In the graph, it is seen that the circuit has a large plateau producing $P(|0\rangle) = 0.5$ where $x > 0$. Given the diminishing step size of the SPSA optimizer, it is possible that a circuit with label $|1\rangle$, that is, where $P(|0\rangle)$ should be minimum, would minimize to $x > 0$ and be stuck at local minima producing $P(|0\rangle) \geq 0.5$.

C. RX-CNOT ANSATZ

The Rx-CNOT Ansatz, meanwhile, has significantly better performance, displaying 96.91% accuracy over the test set for the non-parameterized nouns version. The training graph also reflects a constantly increasing training accuracy, implying that the Rx-CNOT model has more potential to converge. This is corroborated by the fact that the circuit output space, depicted in Fig. 17. In either direction, there are no

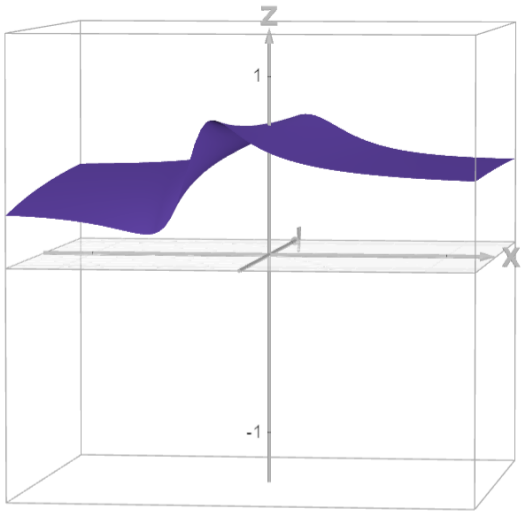


FIGURE 16. H-CRx circuit output space. The x- and y-axes plot the values of θ_3 and θ_4 adjective parameters respectively, and the z-axis represents the probability of measuring $|0\rangle$ for the given parameter values.

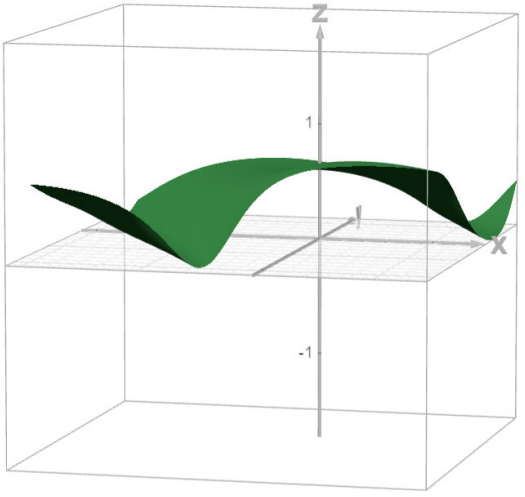


FIGURE 17. Rx-CNOT circuit output space. The x- and y-axes plot the values of θ_3 and θ_4 adjective parameters respectively, and the z-axis represents the probability of measuring $|0\rangle$ for the given parameter values.

plateaus, and the model can comfortably converge to the local maxima for $|0\rangle$ sentences and the local minima for $|1\rangle$ sentences.

An inspection of the Rx-CNOT model's outputs show that all incorrectly labeled sentences are cases where the output of the circuit approaches a uniform superposed single-qubit state, *i.e.* $P(|0\rangle) = P(|1\rangle) = 0.5$. Those sentences are displayed in Table 5. Note that the table lists all the wrongly predicted sentences of the Rx-CNOT model. In fact, all the incorrectly labeled sentences fall into one of two categories: SXA with the adjective "slow", and SVAO with the verb-adjective pair "had yummy". In other words, they involve the $n^r_{slow} \cdot s_{slow}$ parameters and the $n_{yummy} \cdot n^l_{yummy}$ parameters. It is unclear why the *slow* and *yummy* parameters are poorly optimized in this training instance. Visualization of the SXA and SVAO circuit output spaces indicate that there are no barren plateaus in the Rx-CNOT model, which suggests the parameters are trainable. Upon comparison to other adjective parameters, it is unlikely that the poor optimization of *slow* and *yummy* is due to bad initialization either, as the $n^r_{awful} \cdot s_{awful}$ parameters of the adjective *awful* and the $n_{tasty} \cdot n^l_{tasty}$ parameters of the adjective *tasty* have similar initial values yet are well-optimized by the first checkpoint (130 iterations).

Further research may utilize regularization methods or learning-rate adjustments to account for outlier parameters such as "slow" and "yummy" in this example to address poor optimization of these parameters. Additionally, further work may investigate alternative optimization algorithms and refine the Rx-CNOT ansatz to achieve more robust parameter

TABLE 5. Incorrect predictions of the Rx-CNOT ansatz.

Sentence	Prediction	Label
The staff was slow	$\begin{bmatrix}0.5068\\0.4932\end{bmatrix}$	$\begin{bmatrix}0\\1\end{bmatrix}$
The waiter was slow	$\begin{bmatrix}0.5072\\0.4928\end{bmatrix}$	$\begin{bmatrix}0\\1\end{bmatrix}$
The restaurant had yummy food	$\begin{bmatrix}0.4963\\0.5037\end{bmatrix}$	$\begin{bmatrix}1\\0\end{bmatrix}$
The restaurant had yummy meals	$\begin{bmatrix}0.4972\\0.5028\end{bmatrix}$	$\begin{bmatrix}1\\0\end{bmatrix}$
The cafe had yummy cooking	$\begin{bmatrix}0.4770\\0.5230\end{bmatrix}$	$\begin{bmatrix}1\\0\end{bmatrix}$
The cafe had yummy dishes	$\begin{bmatrix}0.4851\\0.5149\end{bmatrix}$	$\begin{bmatrix}1\\0\end{bmatrix}$

TABLE 6. Incorrect predictions of the Rx-CRz ansatz.

Sentence	Prediction	Label
The restaurant showed great service	$\begin{bmatrix}0.4178\\0.5822\end{bmatrix}$	$\begin{bmatrix}1\\0\end{bmatrix}$
I had great food	$\begin{bmatrix}0.4107\\0.5893\end{bmatrix}$	$\begin{bmatrix}1\\0\end{bmatrix}$
The restaurant had great staff	$\begin{bmatrix}0.4213\\0.5787\end{bmatrix}$	$\begin{bmatrix}1\\0\end{bmatrix}$
The cafe showed great foodservice	$\begin{bmatrix}0.4020\\0.5980\end{bmatrix}$	$\begin{bmatrix}1\\0\end{bmatrix}$
I had great cooking	$\begin{bmatrix}0.4306\\0.5694\end{bmatrix}$	$\begin{bmatrix}1\\0\end{bmatrix}$
The cafe had great waiters	$\begin{bmatrix}0.4053\\0.5947\end{bmatrix}$	$\begin{bmatrix}1\\0\end{bmatrix}$

training results. From a practical standpoint, the Rx-CNOT ansatz without noun parameterization has faster fit times compared to the standard IQP ansatz and therefore can be employed in real-world QNLP sentiment analysis pipelines to mitigate for long training times due to sentence size.

D. Rx-CRz ANSATZ

As the Rx-CRz ansatz has two parameterized layers, an adjective parameter in an SVAO sentence (composed of two grammar types) will have 2×2 variables. Consequently, a mapping to the 3D plane is not possible, and a straightforward graphical analysis cannot be employed to determine the features of the Rx-CRz circuit output space.

VI. CONCLUSION

In conclusion, the GeneralQC ansatz achieves higher accuracy than SimpleSA ansatz at sentiment classification. Specifically, it is the Rx-CNOT and Rx-CRz constructions that are significantly more accurate at classifying sentence sentiment compared to SimpleSA ansatz and the H-CRx construction of the GeneralQC ansatz. Both the Rx-CNOT and Rx-CRz constructions achieve 96.91% accuracy, exhibiting over a 15-point increase in accuracy over the previous SimpleSA ansatz. Circuit analysis shows that the Rx-CNOT construction is also inherently better than the H-CRx construction as it does not have areas where the output space is limited to one output. In terms of consistency, the Rx-CNOT model has similar performance on both SVAO and SXA sentence types, whereas the Rx-CRz model performs perfectly on SXA sentences and slightly worse than Rx-CNOT on SVAO sentences. Both circuits are robust to noun choice due to the noun non-parameterization strategy inherited from the SimpleSA ansatz.

In the future, more ansatz constructions may be explored and compared to the constructions offered in this paper. The development of VQA methods is anticipated to improve as more task-specific ansätze are developed, as done in this study. For the task of sentiment classification, a softer tuning method can be used to limit the influence of nouns, such as a rotation gate with restricted parameters. For more specific tuning, it is also possible to extend the strategy to other parts-of-speech and model each word box differently depending on their hypothesized semantic function within a sentence. Additionally, further work can be done to verify the results of this study by performing optimization on noisy devices.

As quantum hardware technology advances, it is expected that complex machine learning tasks may be delegated to quantum computers. This work expands the foundations of quantum natural language processing so that when sufficient device capabilities become available, large-scale implementations of QNLP may be readily tested and developed for industrial use. The use of task-specific ansatz designs such as the GeneralQC ansatz is also intended to inspire ansatz designs in other NLP tasks outside of sentiment classification.

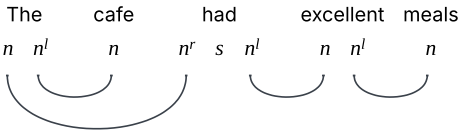


FIGURE 18. Parsed form of the sentence "the cafe had excellent meals."

APPENDIX

The steps from Fig. 3 are explained in this appendix. An example of sentence-to-circuit transformation using the sample sentence "the cafe had excellent meals" is given as follows.

To begin, the sentence is parsed using a grammar parser. This example uses the Bobcat parser provided in Lambeq's latest version (0.4.3). As "cafe" and "meals" are both nouns, they are assigned the n type. The determiner "the" and the adjective "excellent" are assigned the $n \cdot n^l$ type to obtain reductions with their respective nouns that produce an n type, indicating a noun phrase. The verb "had" is therefore assigned the $n^r \cdot s \cdot n^l$ type to obtain a reduction to the s type, indicating the sentence "the cafe had excellent meals" is a syntactically complete sentence. The result of parsing is shown in Fig. 18.

The parsed sentence is then encoded onto a string diagram, as shown in Fig. 19. As a graphical representation of quantum information, the word boxes represent operations while the wires represent the flow of information. The open s wire of the "had" box contains the final state of the sentence. After the string diagrams of all sentences in the dataset are obtained, the dataset is filtered for invalid sentences. Invalid sentences are defined as sentences that do not reduce to a singular s type, *i.e.* they are not syntactically complete sentences. In this example, the types reduce to s, and the pipeline continues.

The diagram is then simplified via rewriting. The term rewriting, as used in this appendix, refers to the process of replacing and transforming boxes or wires on the string diagram to produce a heuristically equivalent string diagram. One such heuristic provided in the Lambeq package are the default rewrite rules. In this example, the determiner rule is applied to rewrite the "the" box out of the diagram. The result of this rewriting is shown in Fig. 20. As this rewrite rule produces a "cap" where the determiner previously was, the yanking equations are applied to normalize the diagram [6]. The resulting diagram is shown in Fig. 21.

Next, the diagrams are rewritten using the bigraph method. This method transposes the boxes of words that are an odd degree away from the root word box – the root word box is the box possessing the open s wire. This allows for the simplification of the associated quantum circuit, as it approximately halves the number of qubits required to implement the diagram. In order to apply the bigraph method, the diagram must first be converted into trailing cup form to allow for the correct order of transposition. The trailing cup

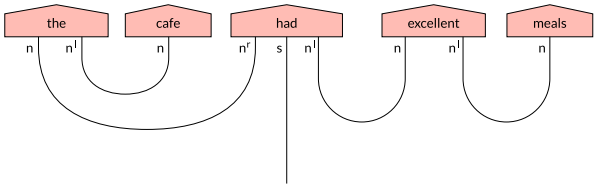


FIGURE 19. String diagram of the sentence "the cafe had excellent meals."

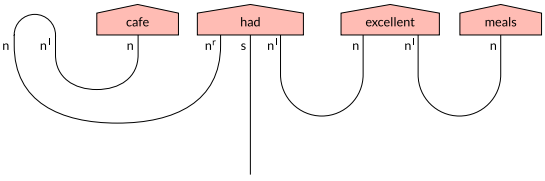


FIGURE 20. Rewritten string diagram of the sentence "the cafe had excellent meals."

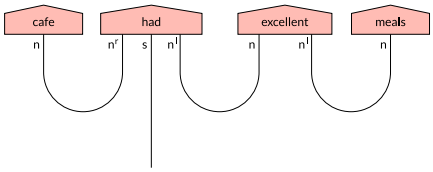


FIGURE 21. Normalized string diagram of the sentence "the cafe had excellent meals."

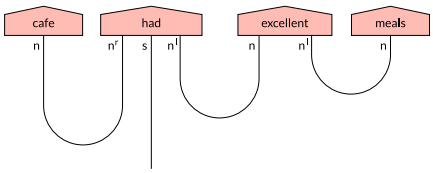


FIGURE 22. Trailing cup form of the string diagram for the sentence "the cafe had excellent meals."

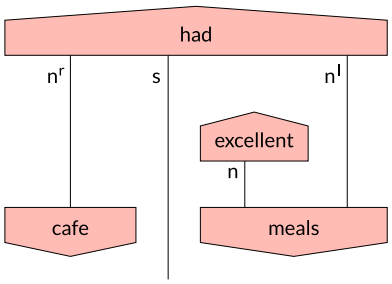


FIGURE 23. Bigraph form of the string diagram for the sentence "the cafe had excellent meals."

form for the example sentence is shown in Fig. 22 while the resulting bigraph diagram is shown in Fig. 23.

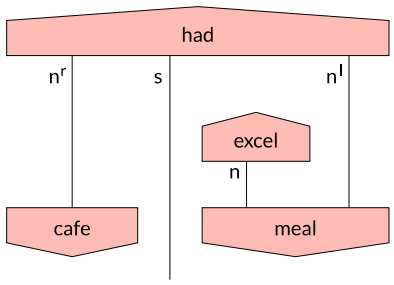


FIGURE 24. Stemmed string diagram of the sentence "the cafe had excellent meals."

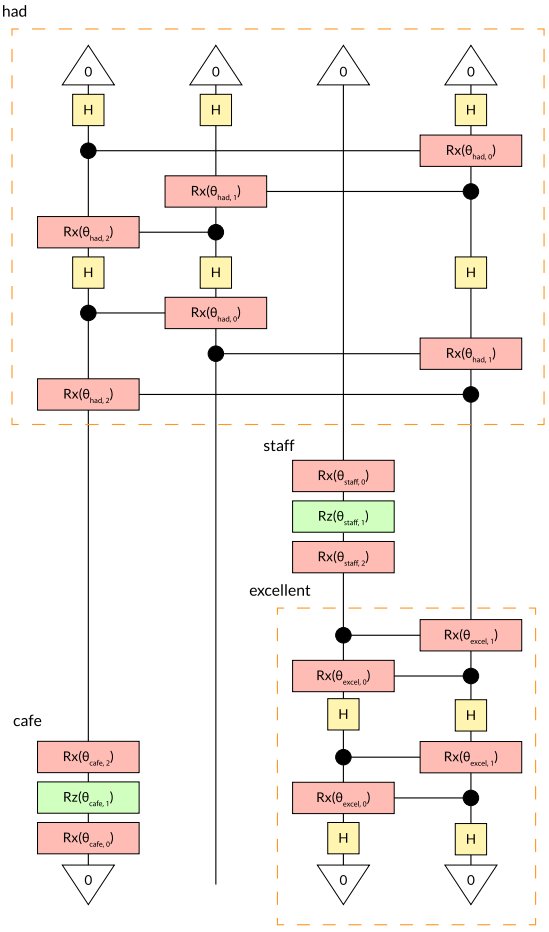


FIGURE 25. Quantum circuit for the sentence "the cafe had excellent meals."

The last rewrite method applied is stemming. Stemming, commonly performed in natural language processing, allows for the reconciliation of words with the same stem but having different morphemes. This is intended to reduce the amount of tokens the model is required to learn, thus speeding up training. The stemmed diagram is shown in Fig. 24.

After obtaining the stemmed diagrams, the sentences are ready to be converted to quantum circuits. Conversion is

done using an ansatz. The ansatz used in this example is the GeneralQC ansatz proposed in this research. The specific construction chosen is the H-CRx construction. The resulting quantum circuit is depicted in Fig. 25. As this sentence does not have the word ''not'', the last step in the circuit processing stage of the pipeline, NOT-box application, is not performed. However were it to be performed, a Pauli-X gate (the NOT gate) would be applied to the end of the second qubit.

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